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Response to Request for Information (RFI) on AI Governance and Safety

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Introduction

Artificial Intelligence (AI) is at a crossroads, with concerns ranging from ethical dilemmas to existential risks. Our approach leverages **Knowledge Graphs (KGs) and Mentored AI with Domain-Specific Trainers** to build an AI system that is **transparent, self-improving, and auditable**, thereby addressing safety concerns while ensuring practical benefits. By integrating **human oversight, logical reasoning, and counterexample-based learning**, our framework offers a **scalable, responsible, and governable** AI model that aligns with societal and governmental needs.

1. AI Governance through Knowledge Graphs and Mentored AI

A. AI Solution Discovery & Storage in Knowledge Graphs (KGs)

1. AI autonomously **discovers solutions** through structured problem-solving using **logical reasoning** and **Prolog-based rule validation**.
2. Solutions are **stored in a domain-specific KG**, ensuring traceability and accessibility.
3. The KG **captures not only solutions but also counterexamples**, allowing for systematic improvement and preventing errors from propagating.

B. Mentored AI with Domain-Specific Trainers

1. AI is trained with **domain-specific experts**, allowing it to develop **industry-aligned knowledge** rather than relying purely on black-box neural networks.
2. AI **debates with itself**, resolving contradictions before presenting solutions for human review.
3. AI undergoes **Socratic questioning** to justify its reasoning, making it accountable and resistant to hallucinations.

C. Human Review and Licensing of AI-Generated Knowledge

1. Solutions generated by AI **require human sign-off** before deployment, ensuring that **humans remain in control**.
 2. AI-generated knowledge can be **licensed** in controlled environments, allowing governments and institutions to regulate and audit AI outputs.
 3. The KG **provides a transparent, queryable structure**, preventing AI from operating in an opaque or uncontrollable manner.
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2. Proof of Concept: Reasoning Language Model (RLM)

A. Language Games as the Foundation of Safe AI

Our **Reasoning Language Model (RLM)** serves as a proof of concept demonstrating that **language games**—rooted in the philosophies of **Aristotle, Socrates, and Wittgenstein**—can form the basis of **safe and explainable AI**. By structuring AI reasoning within logical, ethical, and philosophical constraints, the RLM ensures AI remains **accountable, interpretable, and grounded in human values**.

B. ARC Puzzle Knowledge Graph Schema

The **ARC Puzzle KG Schema** provides a structured approach to solving AI-driven logic puzzles, demonstrating the RLM in action:

1. The schema begins at the **"ARC Puzzle"** node, representing the fundamental problem.
2. Each puzzle is described through a set of **task descriptions**, defining the input-output relationship.
3. **Seven AI Agents** work independently on their own **transformations**, applying different logical and pattern-based solutions.
4. Each agent evaluates its own transformation against the expected outcome:
 - If the solution is correct, it is labeled **"AI Solved"** in the KG.
 - If incorrect, it is labeled **"AI Counterexample"** and stored for further refinement.
5. A **human reviewer** can analyze the steps, review the reasoning behind the AI's decision-making process, and either **approve or disprove** the solution.
6. All reasoning steps are **logged and auditable**, ensuring that AI-generated solutions are transparent and can be traced back to their logical foundations.

By following this structured framework, the RLM ensures that AI's problem-solving ability is **explainable, iterative, and continuously improving**, making AI decisions **both accountable and justifiable**.

3. Addressing AI Safety & the Doomsday Narrative

A. Counterexamples as an AI Safety Mechanism

- Instead of blindly trusting AI outputs, our system actively **searches for contradictions and failure points**.
- Monte Carlo simulations, logical rule validation, and counterexample discovery prevent

erroneous AI conclusions from being accepted.

B. Regulatory Oversight through Transparent AI Systems

- Knowledge Graphs allow regulators to **trace AI reasoning and decision-making**, ensuring compliance with ethical and legal frameworks.
- AI-generated solutions are **not proprietary black-box models** but auditable and licensed AI knowledge structures.

C. AI as a Tool for Humanity, Not a Threat

- Our approach **keeps AI subservient to human expertise and validation**, mitigating risks of uncontrolled AGI emergence.
- Unlike AI systems controlled by private entities or moguls like Elon Musk, our framework ensures **distributed, public, and auditable AI governance**.
- The system is designed for **public-good licensing**, preventing monopolization of AI knowledge.

4. How This Framework Ensures a Safe Future for AI

1. **Human-in-the-loop AI:** AI-generated knowledge is **always supervised and validated** before deployment.
2. **Regulatory-Ready Knowledge Graphs:** AI decisions are stored in **queryable, structured databases**, allowing policy-makers to enforce transparency.
3. **Structured AI Ethics:** AI adheres to **predefined ethical frameworks** with logical justifications and counterfactual testing.
4. **Resilience to AI Hallucination & Bias:** AI outputs are cross-checked with **structured logic and adversarial counterexamples**.
5. **Prevention of AI Monopolization:** AI-generated knowledge can be **publicly licensed and reviewed** to prevent corporate monopolies from controlling AI systems.

Conclusion

AI governance must be based on **transparency, accountability, and structured reasoning**. Our **Knowledge Graph and Mentored AI** approach offers a practical, scalable, and safe AI governance model that aligns with public interests and regulatory needs. By leveraging **structured validation, human oversight, and domain-specific mentorship**, this framework prevents AI from evolving into an ungovernable entity while maximizing its potential for societal benefit.

We look forward to contributing to the development of a robust AI Action Plan that prioritizes **human control, transparency, and ethical AI deployment**.

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